

Integrated Continuous Production of Recombinant Therapeutic Proteins

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ABSTRACT: In the current environment of diverse product pipelines, rapidly fluctuating market demands and growing competition from biosimilars, biotechnology companies are increasingly driven to develop innovative solutions for highly flexible and cost-effective manufacturing. To address these challenging demands, integrated continuous processing, comprised of high-density perfusion cell culture and a directly coupled continuous capture step, can be used as a universal biomanufacturing platform. This study reports the first successful demonstration of the integration of a perfusion bioreactor and a four-column periodic counter-current chromatography (PCC) system for the continuous capture of candidate protein therapeutics. Two examples are presented: (1) a monoclonal antibody (model of a stable protein) and (2) a recombinant human enzyme (model of a highly complex, less stable protein). In both cases, high-density perfusion CHO cell cultures were operated at a quasi-steady state of $50\text{--}60 \times 10^6$ cells/mL for more than 60 days, achieving volumetric productivities much higher than current perfusion or fed-batch processes. The directly integrated and automated PCC system ran uninterrupted for 30 days without indications of time-based performance decline. The product quality observed for the continuous capture process was comparable to that for a batch-column operation. Furthermore, the integration of perfusion cell culture and PCC led to a dramatic decrease in the equipment footprint and elimination of several non-value-added unit operations, such as clarification and intermediate hold steps. These findings demonstrate the potential of integrated continuous bioprocessing as a universal platform for the manufacture of various kinds of therapeutic proteins.

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Introduction

Process intensification through conversion from batch to continuous manufacturing has long been applied in other industries, including steel casting (Tanner, 1998), petrochemical, chemical, food, and pharmaceutical (Anderson, 2001; Fletcher, 2010; Laird, 2007; Reay et al., 2008; Thomas, 2008). Despite the differences between these industries, the advantages of continuous manufacturing are always the same, including steady-state operation, small equipment size, high volumetric productivity, streamlined process flow, low cycle times, and reduced capital cost (Utterback, 1994). One example is the ongoing project at the Novartis – MIT Center for Continuous Manufacturing that targets a holistic redesign of the pharmaceutical manufacturing process to achieve fully integrated, continuous flow (Bisson, 2008; Schaber et al., 2011).

According to Reay et al. (2008), the reasons for the relatively slow penetration of new processing methodologies include low tolerance to risk, management concerns about implementing new technologies, over investment in facilities designed for the old principles, the legacy effect of fully depreciated production plants, and, not the least, regulatory constraints. While these impediments to change are particularly strong in the field of biopharmaceuticals, the evolving competitive business environment is incrementally driving biotech industry towards a tipping point where existing barriers may be counterbalanced by the need for radically improved bioprocessing. In all cases, the success of

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the introduction of innovative approaches depends not only on sound technical vision but also on broad support from the entire organization as a matter of corporate strategy (Shott, 2007; Utterback, 1994).

In the field of biotechnology, the overarching business drivers for shortest development times and cost control under stringent quality/regulatory requirements dominate the industry. These factors have predictably resulted in a conservative approach regarding implementation of new technology. Often, only incremental improvements have been adopted over time, while more significant, stepwise technological changes have become less common. As a result, innovation in biotechnology has become more product- than process-centered. While the aforementioned business drivers continue to dominate, there are several emerging factors that increasingly challenge industry conservatism. At present, biotech companies need to flexibly accommodate both large- and small-volume drugs (niche or orphan drugs), preferably within the same manufacturing facilities. The same is true for therapeutic proteins of different nature, e.g., stable proteins, such as antibody, and highly complex (less stable) proteins, such as recombinant human enzymes (rhEnzymes) or hematology factors. Additionally, current biotech companies require rapid adjustment of production capacity to accommodate fluctuating market demands (Kamarck, 2006). Mounting pressure to reduce cost is further catalyzed by the growing competition from biosimilars. As a result, several authors have voiced their concerns about the long-term viability of the traditional manufacturing facility model based on

multiple 10–20 kL batch bioreactors and downstream trains involving chromatographic columns of significant size (Matthews, 2009; Rutter, 2008). While such facilities can handle large-volume drugs and stable proteins, their ability to readily accommodate diverse biologics pipelines or fluctuating market demand is questionable. Furthermore, the capital investment in building such plants is enormous, which is a major concern if the same model is to be followed when capacity expansion is desired.

This work explores a novel process technology that is meant to address the need for flexibility, consistent product quality, high process output and low cost. The primary intent is to introduce a universal biomanufacturing platform, capable of handling various types of protein drugs (high or low volume, stable or unstable), while significantly reducing the size of the manufacturing equipment and capital cost. Parallel objectives include high operational flexibility, process train simplification through designing out non-value-added unit operations, and streamlining the development efforts through the elimination of lengthy phases from the typical process development timeline, such as scale up and technology transfer. The inclusive list of categorized design goals is presented in Table I.

A key concept of the proposed approach is continuous, steady-state processing that extends downstream of the bioreactor to include the capture step. To date, continuous processing in biotech has focused exclusively on upstream applications. Specifically, continuous (perfusion) bioreactor operation at high cell density has been demonstrated

Table I. Design objectives and related enabling concepts for the integrated continuous bioreactor-capture system.

Category	Design objectives
General	Universal platform for the manufacture of any class of therapeutic proteins Closed system operation enabling the parallel production of different drugs in the same area of reduced environmental class (gray space) High volumetric productivity through continuous flow operation and process intensification Less manual operations and subjective decisions through high level of automation Integrated operation as opposed to disjointed set of unit operations requiring intermediate storage after each step
Quality	High and consistent product quality through operation at optimal steady state Lower level of impurities in the bioreactor due to high cell viability Improved product quality through dramatic decrease of product hold time by minimization of Residence times and elimination of multiple non-value-added hold steps Compatibility with QbD and PAT methodologies Less intermediate testing and stability studies through elimination of hold steps Minimized bioburden risk through closed system operation
Cost	Lower capital cost through significant reduction of facility footprint (small bioreactors and capture columns; elimination of harvest and other intermediate hold tanks) Lower buffer volume and chromatography resin through continuous capture technology Reduced QC environmental testing cost due to closed system operation in a lower environmental class
Speed	Fast development and transfer by preserving the same scale in Late Stage Development, Clinical Production and Manufacturing Streamlined process development through elimination of non-value-added process development phases (e.g., scale up, optimization of intermediate hold steps, intermediate stability studies) Accelerated technology transfer, training and validation through standardization of operating procedures across products
Flexibility	Flexibility, mobility due to significantly smaller equipment size Compatibility with disposable technology Rapid capacity increase/decrease through “numbering up” (through addition of parallel production lines) as opposed to traditional volumetric scale up Low cycle times and simplified logistics through continuous flow operation Simplified transfer to new facility through modularity, standardization and small equipment footprint

repeatedly. These applications have achieved significant improvements in cell banking (Heidemann et al., 2010; Tao et al., 2011), inoculum expansion (Heidemann et al., 2002) and, most importantly, protein production (Ahn et al., 2008; Konstantinov et al., 2006; Ryll et al., 2000; Voisard et al., 2003).

Recently, downstream continuous bioprocessing has become an object of growing interest (Bisschops et al., 2009; Grabski and Mierendorf, 2009; Holzer et al., 2008; Jagschies and Lacki, 2010). In our view, the true potential of continuous chromatography can only be realized by integration with a continuous upstream process (Konstantinov, 2010; Konstantinov et al., 2006; Vogel et al., 2002; Warikoo, 2011). In addition to the aforementioned advantages specific to continuous capture, integration of upstream and downstream unit operations results in the significant simplification of the entire process train due to elimination of non-value-added hold steps, dramatically shorter residence and cycle times, reduction of equipment size, and overall facility minimization. This work describes the successful long-term operation of such a continuous system involving a high cell density bioreactor and a fully integrated chromatographic capture step. Examples of the production of both stable and unstable proteins expressed in CHO cells grown in chemically defined media are provided. Because continuous bioreactor operation has been thoroughly reported in the literature, the greatest focus is given to the integrated continuous capture step.

The impressive successes of continuous flow processing in the pharmaceutical industry have attracted the attention of the regulatory authorities, who are becoming increasingly supportive of this manufacturing paradigm. FDA has recently provided guidance about the applicability of continuous manufacturing of synthetic drugs with highly encouraging conclusions (FDA, 2011; Godwin, 2011). As the parallels to biotech are obvious, these discussions provide a path forward to address the regulatory hurdles for the implementation of the novel continuous bioprocessing paradigm.

Materials and Methods

Cell Culture

Bioreactors with a working volume of 12 L (Broadley-James Corp., Irvine, CA) were operated in perfusion mode utilizing the ATF (Refine Technologies, Pine Brook, NJ) cell retention system with polyethersulfone 0.2 μm filters. Sintered spargers (20 μm) were used for O_2 gas to maintain the dissolve oxygen (DO) set point, and drilled-hole spargers (990 μm) were used for N_2 gas to maintain the pCO_2 set point. Cell density was monitored by offline measurements (Vi-CELL, Beckman Coulter, Brea, CA) and/or via online capacitance probes (Futura, Aber Instruments, Grand Island, NY).

The bioreactor experiments utilized chemically defined media and proprietary CHO cell lines producing monoclonal antibody (MAb) or rhEnzyme proteins. Inoculation viable cell density was 0.5×10^6 cells/mL. The cells were allowed to grow to $50\text{--}60 \times 10^6$ cells/mL, when a cell bleed was initiated to maintain cell density at steady state. Perfusion began 24 h post inoculation at one reactor volume/day with the rate increasing proportional to cell concentration. A steady-state cell-specific perfusion rate of 0.04–0.05 nL/cell-day was maintained. DO was kept above 30% of air saturation. pH was maintained above 6.8 through sodium carbonate addition, but not exceeding 6.95. Antifoam (FoamAway, GIBCO, Grand Island, NY) was used to control foam levels. The harvest obtained from the bioreactors was directly loaded onto the periodic counter-current (PCC) system without additional clarification.

Downstream

The implementation of every new process methodology requires a well-developed hardware base. Unfortunately, the off-the-shelf continuous unit operations developed for the chemical and pharmaceutical industry by companies such as Alfa-Laval, Corning, and Ehrfeld (Hohnson et al., 2008; Mansell, 2008; Short, 2008; Taghavi-Moghadam et al., 2001) are not directly applicable to biotech. To transfer the key principles, new or modified unit operations that suit the specifics of the biological system are necessary. Recently, several continuous chromatography systems have been made commercially available by Novasep (Pompey, France), Tarpon (Worcester, MA), Semba (Madison, WI) and GE Healthcare (Piscataway, NJ), which open up novel opportunities for the implementation of the integrated continuous bioprocessing concept. Some of these units provide options for closed operation using disposables or a SIP approach, which is desirable for robust, long-term performance with the added advantage of reduced environmental classification.

PCC Chromatographic System

The PCC system used in this study was a custom modified ÄKTA (GE Healthcare) capable of running up to four columns. The system was equipped with five UV monitors (UV-900), three pumps (P-900), multiple valves (PV-908, SV-903), one pH and one conductivity meter (pH/C-900), and a Unicorn-based custom software (GE Healthcare).

Breakthrough Curves

Protein breakthrough curves are required to determine the appropriate column switching strategy for PCC. To obtain breakthrough curves under the capture conditions, frontal loading experiments were performed (Cazes, 2001). The dynamic binding capacity (DBC) was evaluated as a

function of residence time (RT) using clarified harvest and breakthrough profiles measured by UV absorbance (280 nm). Column sizes of 6.0 and 1.0 mL were used for the determination of DBC for MAb and rhEnzyme, respectively. The RTs were selected such that they were sufficiently long to satisfy the binding capacity requirements, while also ensuring that the loading time was longer than the rest of the column operations (wash, elution, regeneration, etc.). Accordingly, the loading flow rate was adjusted to achieve a target RT of 2.5 min for MAb and 4.8 min for rhEnzyme.

Integration of PCC to the Bioreactor

In PCC, the RT of the protein on the column can be decreased without increasing the column size because the breakthrough from the first column can be captured on the second column. This unique feature was used to design a continuous process such that the culture harvest could be processed at any perfusion rate (D) by varying the column volume (V) and RT, as outlined by Equation (1):

$$V = D \times RT \quad (1)$$

To achieve continuous capture of the recombinant protein, the PCC was directly connected to the bioreactor as shown in Figure 1. The harvest from the bioreactor/ATF was pumped into a 2 L disposable bag serving as a small surge vessel (Hyclone, Logan, UT) using a peristaltic pump (Masterflex, Cole-Parmer, Vernon Hills, IL). A 0.2 μ m filter (Millipack 40, Millipore, Billerica, MA) was added between the bioreactor and the surge bag as an additional

sterile barrier. MabSelect SuRe (GE Healthcare) and a Hydrophobic Interaction Chromatographic (HIC) media in a XK16TM, 1.6 cm $\times$ 6 cm (GE Healthcare) column was used to capture MAb and rhEnzyme, respectively. Each column operation consisted of equilibration, load, wash, elution, and regeneration steps. Since the engineering of the bench-scale PCC did not allow for closed operation, sodium azide was added to the process stream in-line. This limitation of the small-scale system can be successfully addressed by proper design of large-scale PCC hardware.

Analytical Methods

MAb

In-house assays were used for the quantitation of titer, host cell proteins (HCP), aggregation, residual protein A, and potency. Titer was measured using a Protein A column (Applied Biosystems, Carlsbad, CA). Residual protein A and HCP were quantitated by ELISA using antigens and antibodies produced in-house. Aggregation was measured by HPLC-SEC using a TSK-GEL, G3000SWXL, 7.8 mm $\times$ 30 cm, 5 μ m column (TOSO HAAS, King of Prussia, PA). MAb potency was measured by an in-vitro cell-based assay.

rhEnzyme Activity Assay

The titer of rhEnzyme in the column load and eluate was determined by measuring the hydrolysis rate of a synthetic substrate linked to *p*-nitrophenol (pNP) (Sigma Aldrich, St. Louis, MO). The samples (25 μ L) were incubated with

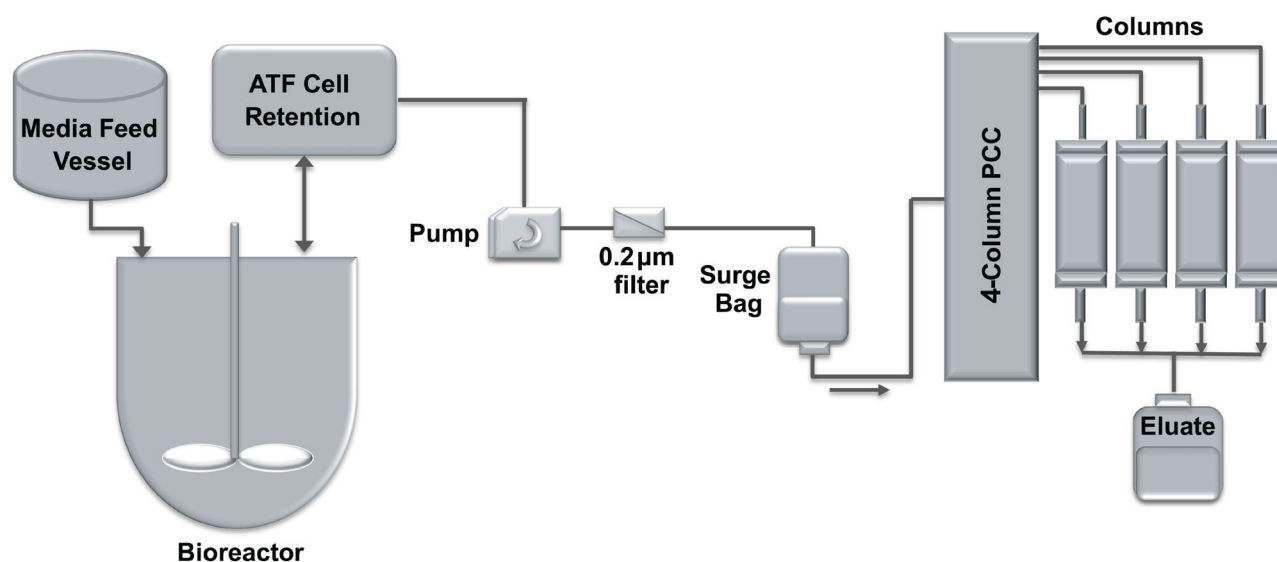


Figure 1. Integration of four-column PCC system with a perfusion cell culture bioreactor to continuously capture the target protein.

225 μ L of 40 μ M substrate for 15 min at 37°C. The reactions were quenched with 250 μ L of 0.3 M glycine, pH 10.5 and the absorbance was measured at 400 nm. One unit of activity was defined as the amount of enzyme required to hydrolyze 1 μ mol of substrate to pNP per minute under the defined assay conditions. Protein concentration was determined by RP-HPLC using a POROS R2/H 2.1 $\times$ 30 mM column (Applied Biosystems). The specific activity was expressed as pNP (units)/milligram protein.

In-house assays were used for the quantitation of HCP, aggregation and purity. HCP was assayed by ELISA using proprietary reagents. The aggregation (SEC-HPLC) assay used a TSK-GEL, G3000SWXL, 7.8 mm $\times$ 30 cm, 5 μ m column (TOSO HAAS, King of Prussia, PA), whereas RP-HPLC purity assay used a YMC Octyl 2 mm $\times$ 100 mm, 5 μ m column (Waters, Milford, MA).

Basic Concepts of PCC

A column operation generally consists of the load, wash, elute, and regeneration steps. In PCC chromatography, multiple columns are used to run the same steps discretely and continuously in a cyclic fashion. Since the columns are operated in series, the flow through and wash from one column is captured by a second column. This unique PCC feature allows for loading of the resin close to its static binding capacity instead of to the DBC as is typical during batch mode chromatography. For the ease of illustration, a three-column system is used to describe the principle of PCC operation (Fig. 2). A cycle is defined as three complete column operations resulting in three discrete elution pools. Once all the steps in a cycle are completed, the cycle is re-started. As a result, the feed stream is processed continuously

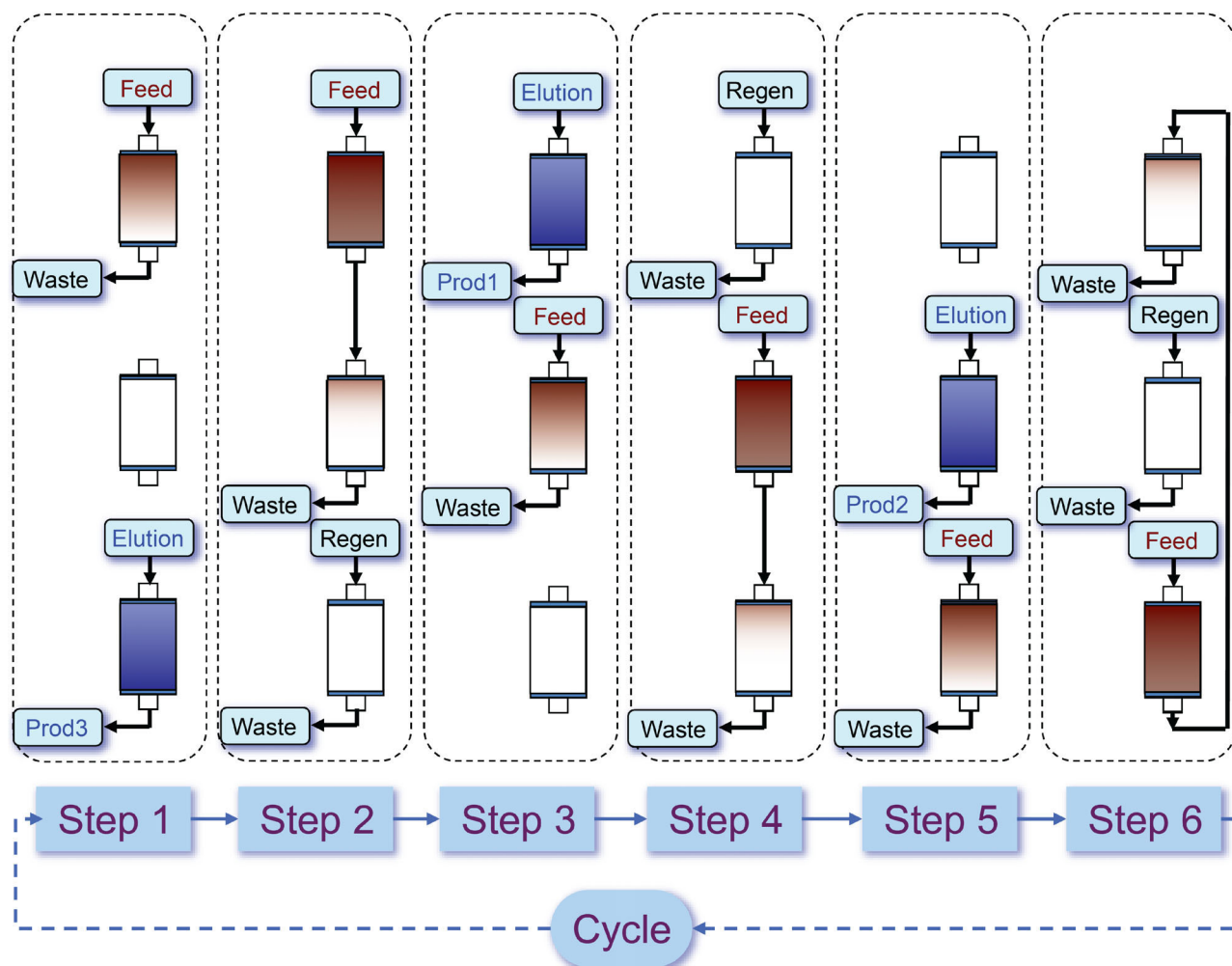


Figure 2. Schematic diagram of the three-column PCC continuous chromatography cycle. At the beginning of a cycle, the feed solution is loaded onto column 1 and the flow-through goes to waste until product breakthrough occurs (step 1). At this point, the flow through from column 1 is directed to column 2 to capture the unbound product from column 1 (step 2). Once column 1 is fully loaded, the feed is now directly loaded onto column 2, while column 1 is washed, eluted, regenerated, and re-equilibrated for the next cycle (steps 3 and 4). Column 2 now goes through steps 3–5, which are identical to steps 1–3 for column 1. Finally, column 3 goes through steps 5–6 in the same way as columns 1 and 2. Once all three columns have completed these steps, the cycle restarts with column 1.

in PCC operation, while product elution from each column is discrete and periodic.

Column Switching Strategy

To advance from one step to another within a PCC cycle (Fig. 2), a column switching strategy is employed. There are two automated switching operations required per column in the PCC, the first of which is related to the initial product breakthrough, while the second coincides with column saturation. The PCC system described in this work was operated using a novel control strategy utilizing dynamic UV monitoring. In general, column switching can be determined by any process analytical technology (PAT) tool capable of in-line measurement of product concentration with feedback control. However, a PAT that operates in real time, such as UV, is ideal for providing the trigger signal for column switching.

Figure 3 shows the principle of column switching based on the UV absorbance difference (ΔUV) between the feed inlet and column outlet. During column loading (step 1; Fig. 2), the PCC control system determines the impurity baseline when the absorbance stabilizes. As the product breaks through (step 2; Fig. 2), there is an increase in the outlet UV signal above the impurity baseline. At the point when ΔUV has reached a pre-determined threshold (such as 3% breakthrough of the product), the flow-through from column 1 is directed onto column 2 instead of to the waste

(t1; Fig. 3). When column 1 is nearly saturated with product and the ΔUV has reached a pre-determined value (t2; Fig. 3), the feed is switched to column 2. An important advantage of this ΔUV -based column switching strategy is that it allows for uniform loading of the columns irrespective of the feed product concentration and the column capacity. Within a reasonable range, the strategy is adequate for harvest titer variability, thereby enhancing system robustness.

As discussed above, accurate determination of the column switching time, which is based on the UV absorbance difference between the feed and column outlet, is one of the critical elements of the PCC real-time control strategy. This requires synchronization of all five UV detectors (one feed and four column outlet detectors) within a narrow range. The UV detectors were calibrated using a 3% acetone solution. The detector path lengths were manually adjusted so that all five absorbance values were within 0.5% of one another. The path length adjustment was $\leq 10\%$.

Results and Discussion

To investigate the broad applicability of the integrated continuous bioprocessing platform, two model proteins with distinct biochemical properties were studied. The model proteins were (1) MAb, representing the class of stable proteins, and (2) a highly complex, relatively less stable non-MAb protein (rhEnzyme).

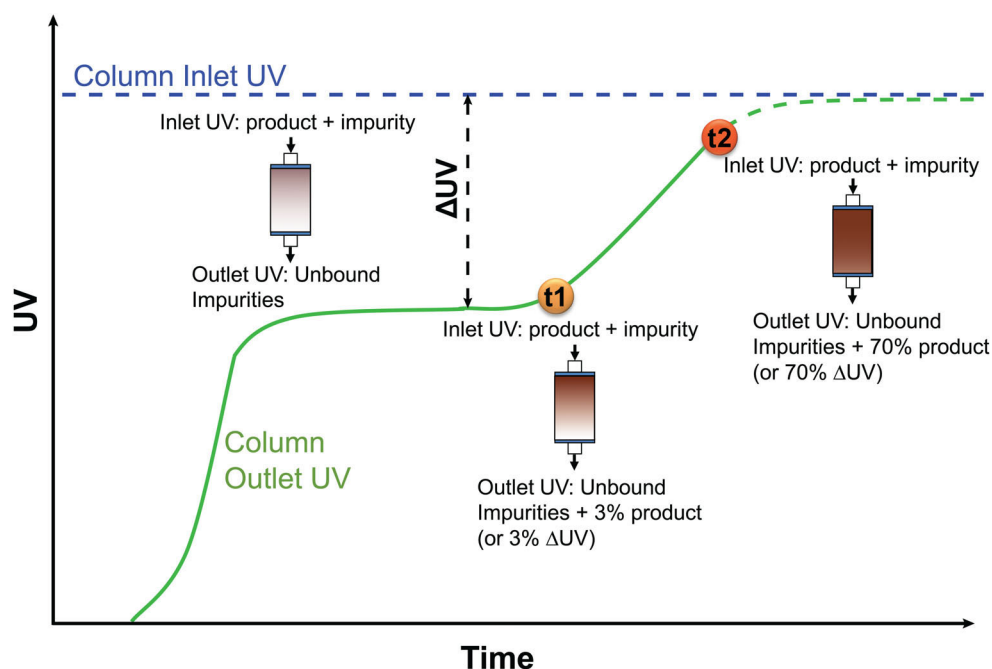


Figure 3. Principle of column switching based on ΔUV . t1 designates the time when the ΔUV has reached a pre-determined threshold. Then, the flow through from column 1 is directed onto column 2 rather than to the waste. t2 denotes the time when the column has been saturated with product. The ΔUV value for both t1 and t2 are process specific.

Proof of Concept using MAb

Cell Culture

The model MAb protein for this study was produced continuously over a 70-day period in a 12-L perfusion bioreactor under the conditions described in Materials and Methods Section (Fig. 4). The volumetric productivity reached 1 g/L-day between days 30 and 40, then slowly declined for reasons that are yet to be determined. The peak volumetric productivity was greater than fivefold than the fed-batch process using the same cell line, with a significant upward potential for the continuous process. It should be noted that no attempts for process optimization were made at this point, as the objective of the study was to demonstrate the functionality of the integrated continuous system. In fact, the volumetric production rate change allowed for the testing of the robustness of the PCC system and, particularly, its ability to handle variability in harvest titer.

Downstream

In batch chromatography, maximum DBC is achieved by increasing the RT and subsequently over sizing the column. In PCC, RT can be decreased without increasing the column size because the breakthrough from first column can be captured on the second column. This advantage leads to smaller column sizes and shorter RT. In order for the process to be continuous, RT has to be longer than the combined time taken by the rest of the process (equilibration, wash, elution, regeneration, etc.). Therefore, the size and number of columns to be used with PCC for a given process are dependent of the resin-binding capacity, which dictates the length of the load step. The high-binding capacity of the MabSelect SuReTM resin in the MAb process (50 g/L) leads

to a load step that is longer than the rest of the column steps combined, which allows for continuous product capture using a three-column PCC process. Frontal loading experiments were utilized to determine the breakthrough curves at different RTs for MAb (data not shown) and an RT of 2.5 min was found to be optimal.

To test the long-term performance of the PCC system, bioreactor harvest was continuously captured for 30 days, which corresponds to 38 cycles and 110 column operations, without any indications of time-based performance decline. The consistency of the continuous capture over the study duration was evaluated based on several performance indicators, such as chromatographic profile, recoveries, and MAb critical quality attributes (CQAs). The UV profile of the feed stream was nearly constant over the duration of the run and the three-column UV outlets were reproducible across various cycles. These details and the cycle-to-cycle reproducibility are demonstrated in the run chromatogram snapshot shown in Figure 5. The recovery and five CQAs analyzed for the capture eluate were comparable between the three columns, as well as over the entire period of continuous harvest capture (Fig. 6). These results demonstrated the feasibility of the direct continuous MAb capture from a perfusion bioreactor, yielding consistent process performance indicators and CQAs over a prolonged period of time.

The three-column PCC process was compared with an existing single column batch chromatography system in terms of the estimated chromatography column footprint and raw material consumption (Table II). Specifically, chromatography media capacity utilization was increased by 20%, buffer usage was reduced by 25% and individual column size was reduced 75-fold, as compared to batch mode. These results further indicate that applying the PCC technology to the MAb process has significant advantages over the traditional batch manufacturing.

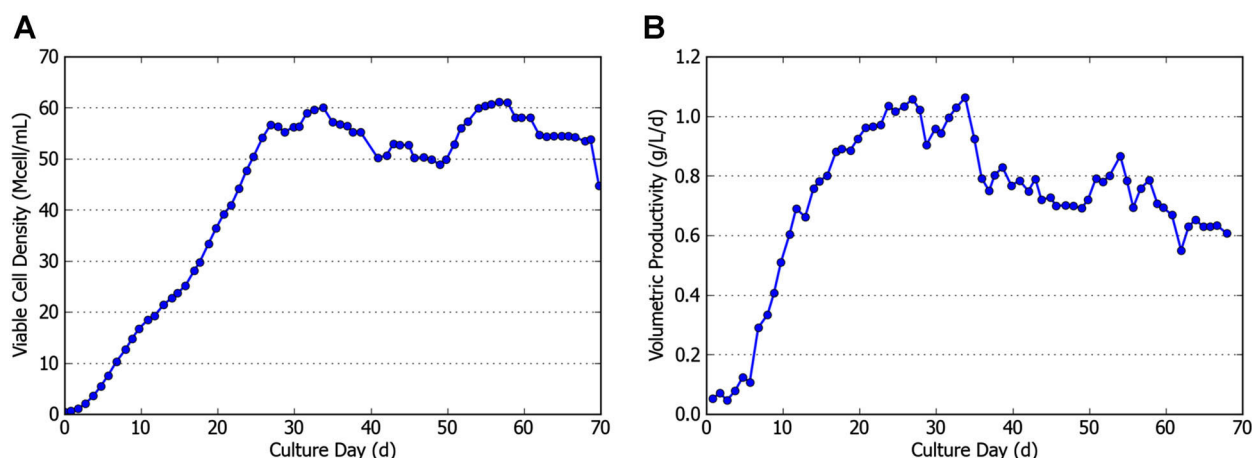


Figure 4. MAb production in a perfusion process: (A) cell density profile in which an average of $50\text{--}60 \times 10^6$ cells/mL were maintained and (B) volumetric production rate.

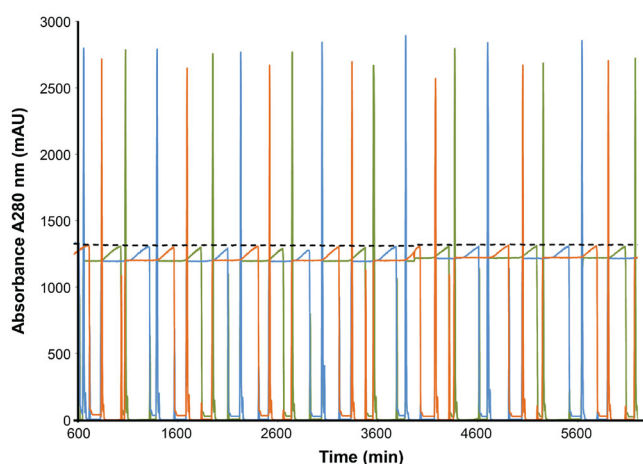


Figure 5. Real-time UV profiles representing the automatic column cycling during continuous MAb capture by PCC. To highlight the details, a limited time window is shown instead of the entire chromatogram for 30 days, 38 cycles, and 110 column operations. Dashed black trace – feed UV absorbance; blue trace – column 1 outlet UV absorbance; brown trace – column 2 UV outlet absorbance; and green trace – column 3 UV absorbance.

Proof of Concept With rhEnzyme

Cell Culture

The rhEnzyme in this study was produced over a 70-day continuous cultivation in a 12-L bioreactor under the conditions described in Materials and Methods Section (Fig. 7). The volumetric productivity reached 0.4 g/L-day around day 25, and remained steady after that with a CV of 16%, largely related to assay variability. Compared to the legacy process for the manufacture of the same molecule, this volumetric productivity is ~40-fold higher, which is a result of the synergistic impact of the high cell density in the

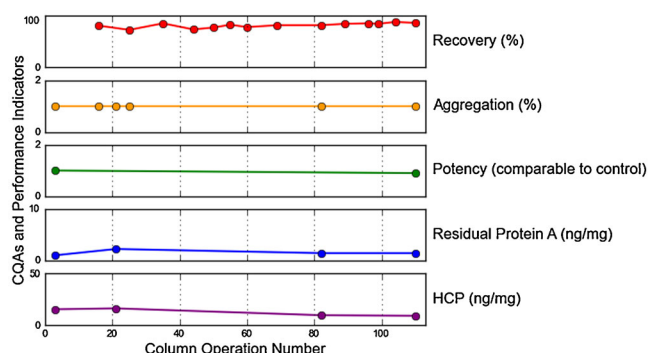


Figure 6. Continuous capture of MAb. Performance indicators and CQAs over 30 days, 38 cycles, and 110 column operations. The variability associated with the three ELISA assays (recovery, residual protein A, and HCP) was $\pm 20\%$. All the results above were within the assay variability.

Table II. Comparison of batch and three-column PCC MAb and four-column PCC rhEnzyme capture.

Parameters	Units	Batch	PCC	
			MAb	rhEnzyme
Resin capacity	Normalized (%)	100	120	150
Buffer usage	Normalized (%)	100	75	54
Column				
Residence time	min	6	2.5	3
Volume	Normalized (%)	100	3.9 (3×1.3)	7.6 (4×1.9)
Diameter	Normalized (%)	100	16	22
Height	Normalized (%)	100	47	40

The values have been normalized to batch mode chromatography for ease of comparison.

new bioprocessing platform and the significantly improved cell-specific production rate.

Downstream

The objective of this case study was to determine the consistency of the capture column performance indicators, CQAs of the capture eluate across the columns and cycles, as well as the robustness of the PCC hardware and control strategy over extended periods of time. Given that the binding capacity of the HIC capture column was low (1 g/L resin), the time required to load a column was shorter than the rest of the process steps combined (wash, elution, regeneration, etc.). This resulted in a PCC method where four columns, instead of three, were required for the harvest to be processed continuously. Therefore, a four-column PCC method was developed and used to continuously capture the rhEnzyme.

The study was divided into two phases. The first phase consisted of method development where previously collected harvest was fed to the PCC from a sterile disposable bag. During the second phase, the PCC was directly integrated with the bioreactor for continuous processing, as outlined in Figure 1. In the first phase of this study, the PCC was operated continuously for 9 days and 41 cycles or 164 column operations. As shown in Figure 8, the UV profile of the feed was constant over the entire run, and the four-column outlet UV signals were reasonably consistent among the four columns and across various cycles. The consistency of the PCC operation during the entire run was demonstrated by the analysis of additional column performance indicators and CQAs of the captured product (Fig. 9). The recovery and the five CQAs analyzed were comparable between the four columns over the period of 9 days of continuous harvest capture.

In the second phase of this study, product was directly captured from a perfusion bioreactor by integrating the PCC with the bioreactor, as shown in Figure 1. The main objective was to demonstrate the performance of the continuous bioprocessing platform with a highly complex, non-MAb protein over an extended period of time. The

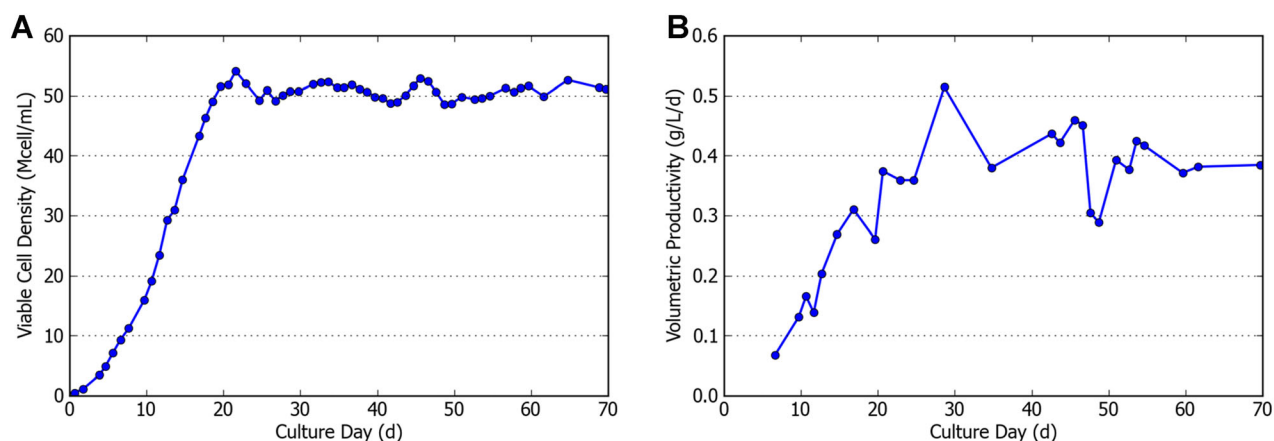


Figure 7. rhEnzyme: (A) cell density profile in which an average of 50×10^6 cells/mL were maintained after Day 19 and (B) associated volumetric production rate (CV = 16% including assay variability). The two low titer values around days 48–49 are believed to be due to assay variability as there were no changes in cell density, perfusion rate, or metabolism over this period.

column size and RT were scaled according to Equation (1) in order to achieve continuous capture of the bioreactor harvest. The integrated system was operated continuously for 31 days and 160 cycles, which corresponds to 640 independent column operations, without any signs of time-based performance decline. The feed UV profile, the four-column outlet UV profiles and CQAs were consistent for the entire duration of the PCC operation. Since the data were practically equivalent to those obtained in the first phase of the study, the time profiles are not shown. Table II summarizes the advantages of the PCC rhEnzyme capture in

comparison to the existing batch process. Specifically, chromatography media capacity utilization was increased by 50%, buffer usage was reduced by 46%, and individual column size was reduced 50-fold, as compared to batch mode. The dramatic decrease in the chromatography media and buffer requirements was partly due to the fact that the batch capture process was not optimized for resin utilization. The gains would likely be comparable to those observed in the MAb process if the rhEnzyme batch operation was optimized for maximum capacity. Nonetheless, the results for the rhEnzyme further

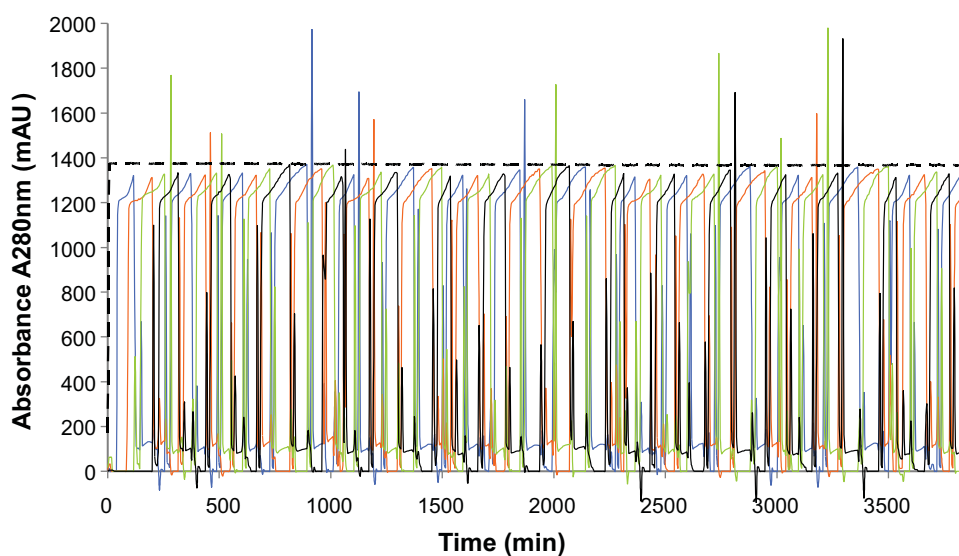


Figure 8. Real-time UV profiles representing the automatic column cycling during continuous rhEnzyme capture by PCC. To highlight the details, a limited time window is shown instead of entire chromatogram for 9 days, 41 cycles, and 164 column operations. Dashed black trace – feed UV absorbance; blue trace – column 1 outlet UV absorbance; orange trace – column 2 UV outlet absorbance; green trace – column 3 UV absorbance; and black trace – column 4 UV outlet.

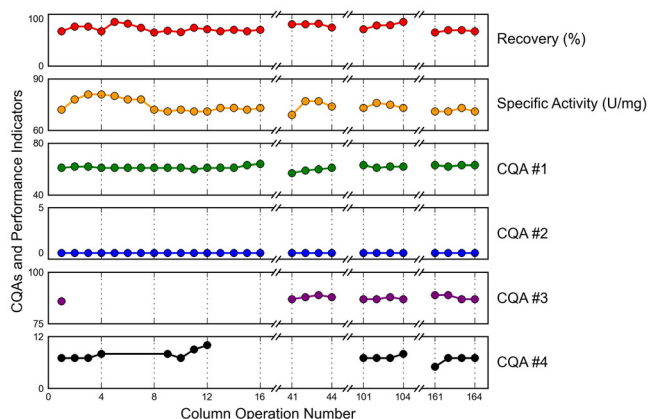


Figure 9. Continuous capture of rhEnzyme. Performance indicators and CQAs over 9 days, 41 cycles, and 164 column operations. CQA # 3 and # 4 have limited data points but represent the entire operational duration. The standard deviation of the CQA's is in the range of 1–9%.

demonstrate the significant advantages of PCC technology when compared to traditional batch manufacturing.

Implementation of the Continuous Biomanufacturing Platform

Currently, there are two dominant platforms for biopharmaceutical manufacturing: (1) perfusion bioreactors,

typically used for production of less stable proteins (Fig. 10, Panel A) and (2) fed-batch bioreactors for production of stable proteins, such as MAbs (Panel B). In both cases, the bioreactor operation is followed by multiple batch unit operations, including clarification, capture, polishing chromatography, and hold steps. The continuous capture technology proposed in this study allows for a significantly streamlined process train (Panel C). As demonstrated in the two examples above, in combination with high producing clones and chemically defined media, this platform can achieve very high cell density and volumetric productivity while operating at steady state. As a result, sufficient production capacity can be achieved with smaller bioreactors (≤ 500 L) versus traditional processes where reactor scales exceed 10,000 L. The use of the ATF cell separation device eliminates the clarification unit operation. Most importantly, the direct integration of the PCC continuous capture step makes harvest hold tanks obsolete and replaces the large batch capture column with up to two orders-of-magnitude smaller columns used in the continuous system. Furthermore, continuous processing of the harvest confers significant advantages with respect to protein quality. Specifically, elimination of the harvest and other hold steps decreases target protein exposure to enzymatic, chemical, and physical degradation and thereby mitigates product stability risks. In summary, the successful development of the continuous bioprocessing platform at small scale for the manufacture of two diverse recombinant proteins opens up the potential for its large-scale industrial implementation.

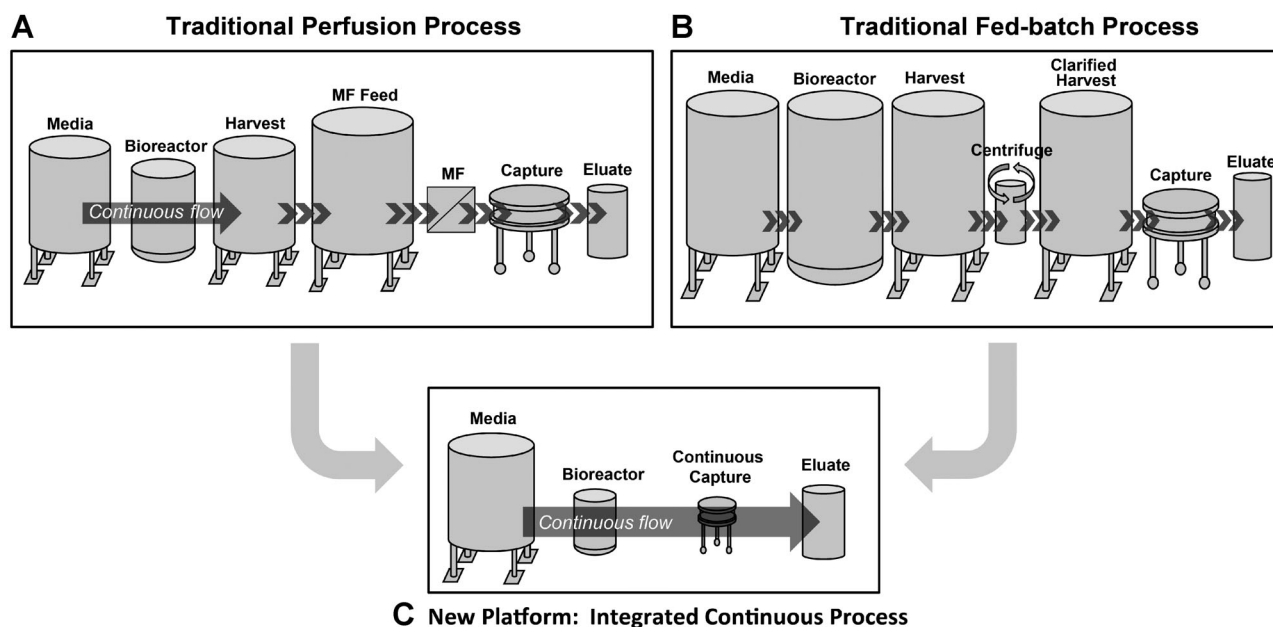


Figure 10. Schematic of traditional and future manufacturing platforms for perfusion bioreactor process. (A) Traditional perfusion manufacturing process; (B) traditional 10–20 kL fed-batch manufacturing process; and (C) new integrated continuous manufacturing platform.

Conclusions

Integrated continuous bioprocessing is a novel solution that offers unique advantages over traditional approaches for recombinant protein manufacturing. This new platform has been successfully applied at development scale to drugs with diverse properties, such as a high-volume stable protein (MAb) and a low-volume, less stable protein (rhEnzyme), which define the boundaries of real-world production scenarios. The large-scale implementation of the platform requires a fully closed PCC system that can operate under sterile conditions over prolonged periods of time. While this requirement appears challenging, it can be accomplished with the engineering toolbox existing today. The flexibility of the system may be further enhanced by incorporating disposable solutions, both upstream and downstream.

Our vision of the biomanufacturing “facility of the future” based on the integrated continuous platform is outlined in Figure 11. This general floor plan utilizes multiple parallel and independent production lines designed as a closed system that offer multiproduct and multiphase manufacturing capability. The flexibility of this scheme

enables rapid increase or decrease of production capacity based on real-time market demand using a “numbering up” approach rather than the traditional volumetric scale up. Large- or small-volume drugs, and the production of either stable or unstable proteins, can be achieved while operating at a high level of standardization and mobility. As the equipment footprint is dramatically smaller, the size of the required manufacturing facility and the related capital cost are significantly reduced.

The proposed bioprocessing platform moves the continuous-to-batch boundary downstream of capture in the process train. The remaining batch purification steps still require large columns, as the transition from continuous to batch is always associated with a major increase in equipment size and, correspondingly, facility scale. The continuous-to-batch transition also requires a change in logistics and an introduction of various hold steps of substantial volume. To avoid these complexities in the long term, extension of continuous operation downstream of the capture step will reduce equipment volume and processing times even further, eventually encompassing the entire biomanufacturing train within an integrated continuous flow. While this proposition will likely face some skepticism,

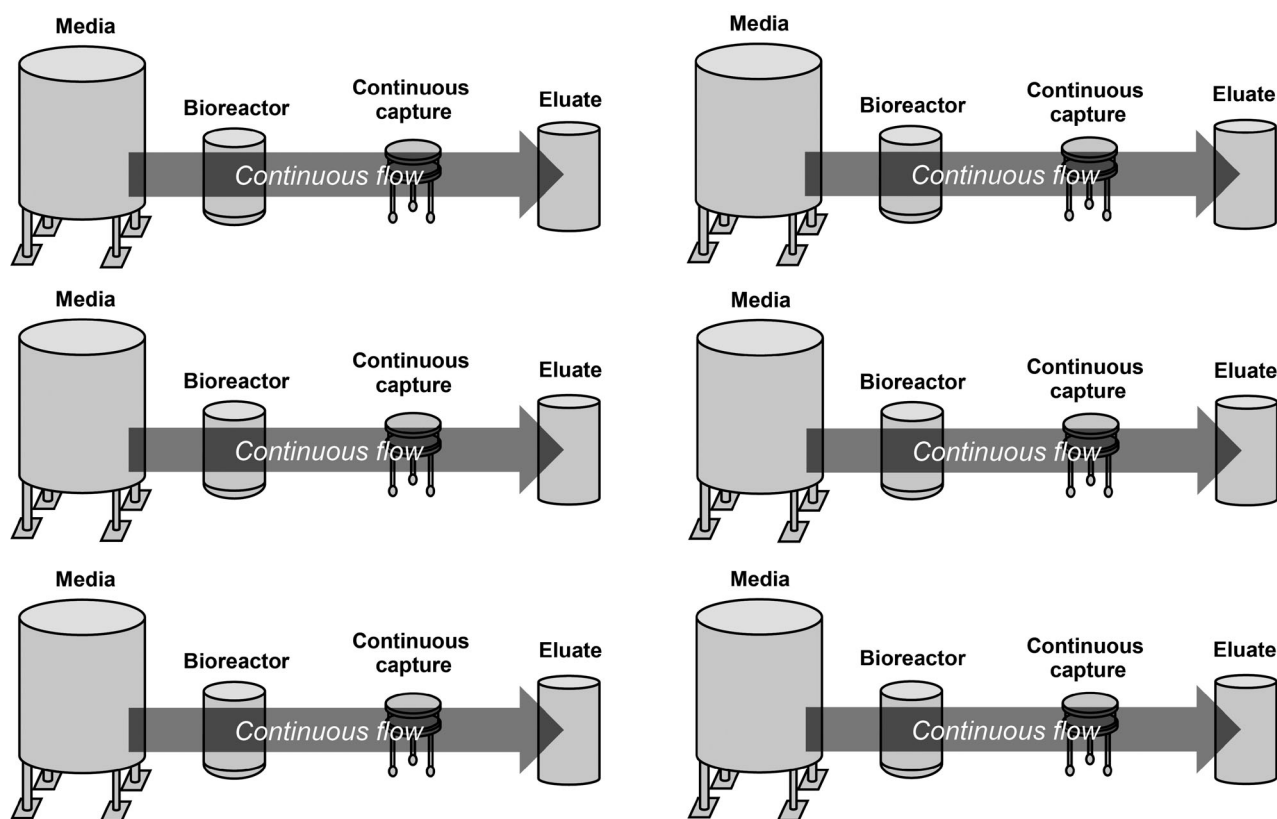


Figure 11. General floor plan of a multiproduct biomanufacturing facility utilizing the integrated continuous bioprocessing platform, implemented as six independent and parallel process trains. The key concepts presented in the text (e.g., closed systems, modularity, small equipment footprint, flexibility, “numbering up” instead of scale up) are essential part of the facility design.

it is important to consider the lengthy but highly successful evolutionary path of the continuous manufacturing paradigm in other industries (Tanner, 1998; Utterback, 1994), which is seen today as a “disruptive technology” that has moved the business to a new utility slope (Christensen, 2000; Levine, 2010).

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